

App Flow

Upaj (Farmer Application)

1. **Home Dashboard:** High-level overview of live field sensors, active crop listings, and summarized monthly net profit.
2. **Weather & Field:** Defaults to regional forecasts. A "Pair Hardware" button triggers BLE handshake with ESP32. Upon connection, renders side-by-side local sensor data (Soil Moisture, Temp) and generates the automated logic advisory.
3. **Disease Detection:** Camera viewfinder interface. User captures leaf image; the app processes the tensor locally and returns a diagnostic card with confidence %, severity, and organic/chemical treatments.
4. **Hyperlocal Market:** Farmer enters crop type and quantity. System disables manual pricing, generates the Agmarknet-backed Smart Price (with a $\pm 5\%$ quality slider), and lists the crop to nearby buyers. Contains native phone dialer links to call interested buyers.
5. **Farm Business Overview:** Data visualization view displaying Net Profit formulas and horizontal ROI progress bars for specific crops.

Mandi (Buyer Application)

1. **Onboarding & Preferences:** Buyer sets business type (Kirana, Restaurant) and sourcing radius slider (e.g., 5km - 25km).
2. **Discover Feed:** List/map view populated by `ST_DWithin` spatial queries. Cards display crop, distance, farmer name, and Smart Price.
3. **Order Negotiation:** Buyer can accept the listed price or counter-offer (enforced within the $\pm 5\%$ boundary).
4. **Direct Handshake:** "Call Farmer" button triggers mobile OS `tel:` scheme for verbal deal finalization.